DEPARTMENT OF CSE

YEARLY QUESTION

ON

ATOMIC STRUCTURE

1.Write down the postulates of Bohr's atomic model. 2 (2015,2013 , 2020

2.Discuss the drawbacks/inadequacies of Rutherford's model for atomic model structure. 1.5/2 (2018,2016,2012

3.State the limitations of Bohr's atomic model. 1.5 (2018 , 2020

4.Explain with examples Hund's rule of maximum multiplicity.How does this help in determining the electronic configuration of elements. 2.75 (2018,

5.Derive an expression for the radius ~~for the radius~~ of n-th orbit of H-atom 3.75

(2019,2017,2016,2014,2012

6.Calculate the radii of fourth orbit in H-atom. Given that $h=6.626\times10^{-27}$ erg-s,$m=9.1\times10^{-28}$ g $\pi=3.14$,$e=4.8\times10^{-10}$ esu

7.Calculate the radius of second orbit He^+ ion 1.5 (2019

8.Calculate the radius of 1st orbit He^+ ion .Given that Bohr radius is 0.529×10^{-10} m 2 (2014,

9.Find out the four quantum numbers for the 15th electron of Cl-atom 1.75(2017

10.State and explain the following term:Aufbau principle,Hund's principle,Pauli's exclusion principle, Heisenberg uncertainty principle . 02 (2019,2018,2017,2016,2013

11.How many subshells are possible in third energy shell?what is the maximum number of electrons that can occupy a d subshell? 2 (2013

12.What are quantum numbers?what is the subshell designation for each of the following cases?

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i)n=2,l=0;ii)n=5,l=1;iii)n=4,l=3 1.75 (2016,2014

13.What is (n+l) rule or Why the last electron k(19) goes to 4s orbital instead of 3d orbital 01(2015Which of the following orbital are possible or not possible 1P,2S,2d,3f 1.75 (2017,2013

14.Write down De-Broglie equation or χ=

15.Calculate the wave length in A^0 of the line in Balmer series that is associated with the drop of the electron from the fouth orbit? $R_H = 109{,}676\ cm^{-1}$ 2.75

16. Calculate the frequency in term of wavelength for second spectral line of lyman series.Given that $R_H = 1.097\times10^7\ m^{-1}$ 1.5(2019,2016

17.Calculate the frequency of electromagnetic radiation emitted by hydrogen atom in the electron transition from n=4 to n=3 . Given that $R_H = 1.097\times10^7\ m^{-1}$ and $h=6.626\times10^{-34}$ Js 2 (2015 , 2020

18.The n quantum number of an atomic orbital is 6.What are the possible values of l? what are the possible value of m_l if l quantum number is 5? 1.75 (2015

19.State whether each of the following sets of quantum numbers is permissible for an electron in an atom.If a set is not permissible, explain why. 1 (2015

(i) (1,0,+1/2,+1/2) and (ii) (3,2,1,1)

20. Write down Rutherford's Experiment of scattering α-particles and give the drawbacks of the model.

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